

RPN Calc

Implement a stack and show it off by writing a programme which reads input from rpnin.txt. After each line of input, print out the contents of the stack, and execute the command given in the table below. If the stack is empty after a pop, print 'stack is empty!'.

+ Pop the top two numbers `__n_1` and `__n_2` off the stack and push $(_n_1 + _n_2)$ on

- Pop the top two numbers `__n_1` and `__n_2` off the stack and push $(_n_2 - _n_1)$ on

* Pop the top two numbers `__n_1` and `__n_2` off the stack and push $(_n_1 \times _n_2)$ on

/ Pop the top two numbers `__n_1` and `__n_2` off the stack and push $(_n_2 / _n_1)$ on

swap Exchange the top two numbers on the stack

dup Duplicate the top of the stack

p Pop the top number off the stack

q Terminate the program, do not print the stack for this line.

otherwise If the input is not a command from above, it is an integer to put on the stack.

You may assume that your stack does not grow beyond 1000 levels deep.

Sample Input

```
1
2
+
4
*
2
/
10
swap
-
dup
p
5
-
p
q
```

Sample Output

```
1
1 2
3
3 4
12
12 2
6
6 10
10 6
4
4 4
4
4 5
-1
stack is empty!
```

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